München

Kai Li

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Kai-Li
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Education

Apr 2024 – Apr 2026 **M.Sc. Computer Science (*Grade: 1.11 / 5.0*)**, *Technical University Darmstadt*, Darmstadt, Germany
Specialization: Machine Learning, Generative AI, and Reliability
Exchange Semester: National Taiwan University (Taipei, Taiwan), Feb 2025 – June 2025

Oct 2020 – Dec 2024 **B.Sc. Computer Science (*Grade: 1.52 / 5.0*)**, *Technical University Darmstadt*, Darmstadt, Germany
Specialization: Natural Language Processing and Artificial Intelligence
Exchange Semester: Högskolan i Skövde (Skövde, Sweden), Jan 2023 – June 2023

Oct 2019 – Sept 2020 **Electrical Engineering (Transfer to CS)**, *Technical University Darmstadt*, Darmstadt, Germany
Completed foundational coursework in mathematics before switching to Computer Science.

Master's Thesis | Fraunhofer IKS & TU Darmstadt

Title **Mitigating and Understanding Hallucinations via Multi-Agent Debate**

Supervisors Josef Jiru (Fraunhofer IKS), Lukas Helff, M.Sc. and Prof. Dr. Kristian Kersting

Description Design and development of a collaborative multi-agent framework to improve the factual reliability and robustness of Large Language Models (LLMs).

- **Model Interpretability:** Developed a detection mechanism based on **internal model activations** using **classification probes** to identify latent errors during the inference process.
- **System Orchestration:** Designed a multi-agent workflow that utilizes iterative "debate" and self-reflection to autonomously verify and refine model outputs.
- **Domain Application:** Validated the framework within a **Requirements Engineering** context, demonstrating significant improvements in consistency and deterministic performance.

Bachelor Thesis | TU Darmstadt

Title **Evaluating Cultural Diversity in Text-to-Image Models**

Supervisors Cecilia Chen Liu, M.Sc. and Prof. Dr. Iryna Gurevych (Grade: 1.0)

Description Architected a large-scale diagnostic benchmark to quantify cultural bias and semantic alignment in Generative Vision-Language models.

- **Data Engineering:** Built and managed a high-throughput pipeline processing **2.3M compositional prompts** across 5 languages to evaluate model representations at scale.
- **Automated Evaluation:** Designed an end-to-end evaluation suite leveraging **Visual Question Answering (VQA)** models to automate the measurement of text-image alignment.
- **Technical Insights:** Quantified systemic performance gaps in non-Western contexts, providing a data-driven basis for improving model robustness and multilingual capabilities.

Research & Engineering Projects

Oct 2021 – Feb 2022 **Neural Sentiment Classifier for Multilingual Tweets**, *TU Darmstadt*

Developed a feedforward neural network for sentiment classification using TensorFlow and NLTK.

- Performed extensive NLP preprocessing (lemmatization, noise filtering) on imbalanced multilingual data.
- Implemented Bag-of-Words vectorization and addressed class imbalance through targeted data strategies.

Leadership & Mentoring

2022 – 2024 **Technical Tutor**, *Department of Computer Science, TU Darmstadt*

Subjects: Language Technology (NLP), Compiler Construction, Visual Computing

- Translated complex theoretical concepts into digestible seminars to improve student engagement.
- Collaborated with academic staff to optimize course delivery and assessment organization.

2021 – 2022 **Student Mentor**, *IMS System, TU Darmstadt*

- Guided first-year students through weekly coaching sessions focused on goal-setting and academic resilience.
- Facilitated self-reflection and growth-mindset workshops for groups of new arrivals.

Technical Skills

Programming	**Python** (Proficient), Java, Bash, PowerShell, LaTeX
AI / Machine Learning	**Transformers**, Pytorch,Interpretability, LLM Fine-tuning, RAG, Computer Vision
Evaluation & Tools	LLM Benchmarking, Manual Error Analysis, Pandas, Matplotlib, Scikit-learn Metrics.
DevOps & Software	**Git**, Linux/Unix, Scrum, Windows, Docker (Basic)

Languages

German	Native	
English	C1 – Advanced	*Professional working proficiency; academic research and publication*
Chinese	B1/B2 – Intermediate	*Heritage speaker; advanced coursework completed at NTU Taipei*
French	A2 – Basic	*Foundational knowledge from secondary education*
Swedish	A2 – Basic	*Introductory coursework completed during exchange at Skövde*

Scholarships & Awards

Feb 2025 – June 2025 **PROMOS Scholarship**, *DAAD / TU Darmstadt*

Awarded for academic excellence to support international research at National Taiwan University.

Oct 2021 – Sep 2023 **Deutschlandstipendium (National Merit Scholarship)**, *Federal Ministry of Education*

Awarded to the top 1% of students based on academic achievement and social commitment.

Interests

Bouldering	Active recreational climber (5+ years); focus on problem-solving and physical endurance.
Global Culture	Passionate about cultural immersion and language acquisition through international travel.
AI Ethics	Keen interest in the socio-technical impacts of generative models and bias mitigation.

Interests

Bouldering	Recreational climber for 5+ years
Traveling	Interested in learning about different cultures and languages
Languages	Actively learning and practicing new languages